know how much stuff is out there and how it behaves. On the other hand, we don't know how to connect either dark matter or dark energy, which together constitute 95 percent of the universe, to fundamental physics. "So in that sense we don't understand at all," he says.

Meanwhile, the view from particle physics is frustratingly similar. Back in the 1970s, physicists developed the Standard Model of particle physics to describe all the known particles of nature: the quarks that make up protons and neutrons, the leptons like the neutrinos and the electron and its cousins, and the so-called gauge bosons that act as go-betweens carrying the fundamental forces between particles (electromagnetism and the strong and weak nuclear forces). Notwithstanding some minor tweaks, like taking neutrinos from being strictly massless to being very, very light, the Standard Model has been fantastically successful, passing every experimental test thrown at it. It even predicted the existence of the Higgs boson—the final piece of the Standard Model puzzle. In the years since, nothing has been discovered in particle experiments that the Standard Model didn't tell us we would find.

You'd think this would be hailed as a triumph. The theory works! Everything is as we predicted!

Why aren't we sitting back and basking in our brilliance and success?

Because this is, in some ways, a worst-case scenario. As great as the Standard Model is at matching experimental results, we know that it, like the Concordance Model in cosmology, has to be missing some very important pieces. In addition to having nothing at all to say about dark matter or dark energy, it has some major "tuning problems"—places in the model where a parameter has to be set juuuuust right or else everything falls apart. Ideally, we should have some theoretical framework that tells us why a parameter is what it is. It's disconcerting when we find that the only reasons we have to set the parameter to that value are "otherwise bad things would happen to us" or, worse yet, "that's just what the measurement says."

For decades, there was hope on the horizon that we might be able to step seamlessly from confirming the important aspects of the Standard Model to finding the edges of its validity and making new discoveries with whatever model we found to replace it. In the 1970s, a model known as supersymmetry (SUSY, for short) was proposed to fix some of the theoretical niggles of the Standard Model by hypothesizing new mathematical connections between different kinds of particles and